Discussion Mini Lecture 1

Math for Machine Learning

Review of Pre-Req Material and Connection to ML

CS 189/289A, Fall 2025 @ UC Berkeley
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Notes about Discussion Mini Lectures

- Discussions are offset from lectures
  - E.g., Discussion 1 takes place in Week 2 and covers concepts presented in Lecture 1 during Week 1
- You can utilize course resources in whatever way works best for you, but we recommend that you:
  1. Attend/watch lectures
  2. Watch the corresponding discussion mini lecture
  3. Attend a discussion section
  4. Review uncovered discussion problems on your own*
  5. Optionally, look at additional resources at end of slides

* We will likely **not** cover all of the discussion material during the 50 min discussion session! Additional problems are provided for you as a way to review concepts and get extra practice.

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Concepts Covered

1. ML as Function Approximation
2. Linear Algebra
3. Multivariate Calculus
4. Probability Theory

ML as Function Approximation

1. ML as Function Approximation
2. Linear Algebra
3. Multivariate Calculus
4. Probability Theory

Supervised Learning

ML Model → dog

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

ML Model → spam

# Bed	# Bath	…	Location
4	3	…	Berkeley

ML Model → $1,200,000

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ML as Function Approximation

f ?

dog

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

g ?

spam

# Bed	# Bath	…	Location
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h ?

$1,200,000

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How do we Represent our Data?

$\longrightarrow$ $\boldsymbol{H} \times \boldsymbol{W} \times \mathbf{3}$ **array of pixels** $\xrightarrow{\textbf{flatten}}$ $\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$ **vector of pixels**

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

$\longrightarrow$ $\boldsymbol{x} = \begin{bmatrix} HI, \\ It's \\ \vdots \\ SOON. \end{bmatrix} \in \mathbb{R}^D$ **vector of words**

# Bed	# Bath	…	Location
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$\longrightarrow$ $\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$ **vector of features**

Practice with data manipulation: Discussion 1 notebook

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ML as Function Approximation

$$\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$$

vector of pixels

$\text{dog} = f(\boldsymbol{x})$

$\boldsymbol{f}$?

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

$$\boldsymbol{x} = \begin{bmatrix} HI, \\ It's \\ \vdots \\ SOON. \end{bmatrix} \in \mathbb{R}^D$$

vector of words

$\text{spam} = g(\boldsymbol{x})$

$\boldsymbol{g}$?

# Bed	# Bath	…	Location
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$$\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$$

vector of features

$\$1{,}200{,}000 = h(\boldsymbol{x})$

$\boldsymbol{h}$?

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Linear Algebra Review

1. ML as Function Approximation
2. Linear Algebra
3. Multivariate Calculus
4. Probability Theory

ML as Function Approximation

$$\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$$

vector of pixels

$\text{dog} = f(\boldsymbol{x})$

$\boldsymbol{f}$?

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$$\boldsymbol{x} = \begin{bmatrix} HI, \\ It's \\ \vdots \\ SOON. \end{bmatrix} \in \mathbb{R}^D$$

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$$\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$$

vector of features

$\$1{,}200{,}000 = h(\boldsymbol{x})$

$\boldsymbol{h}$?

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Linear Function Approximation

$$\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$$

vector of pixels

$$\text{dog} = f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$$

$$\boldsymbol{A} \in \mathbb{R}^{K \times D}$$

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

$$\boldsymbol{x} = \begin{bmatrix} HI, \\ It's \\ \vdots \\ SOON. \end{bmatrix} \in \mathbb{R}^D$$

vector of words

$$\text{spam} = g(\boldsymbol{x}) = \boldsymbol{B}\boldsymbol{x}$$

$$\mathbf{B} \in \mathbb{R}^{K \times D}$$

# Bed	# Bath	…	Location
4	3	…	Berkeley

$$\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$$

vector of features

$$\$1{,}200{,}000 = h(\boldsymbol{x}) = \boldsymbol{C}\boldsymbol{x}$$

$$\mathbf{C} \in \mathbb{R}^{K \times D}$$

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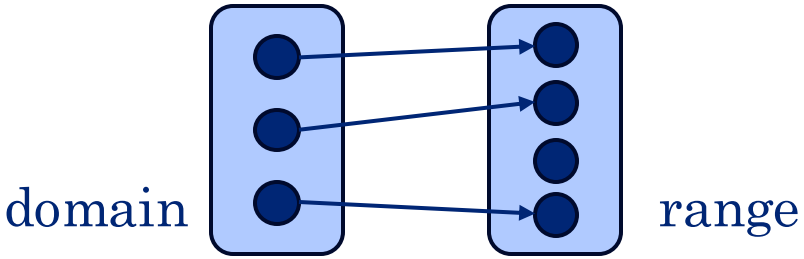
Properties of Functions

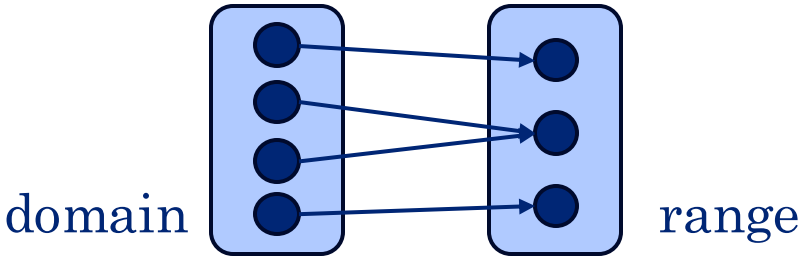
* If an input maps to more than one output, we do not have a valid function.

injective – every input in our *domain* maps to **one*** output in our *range*



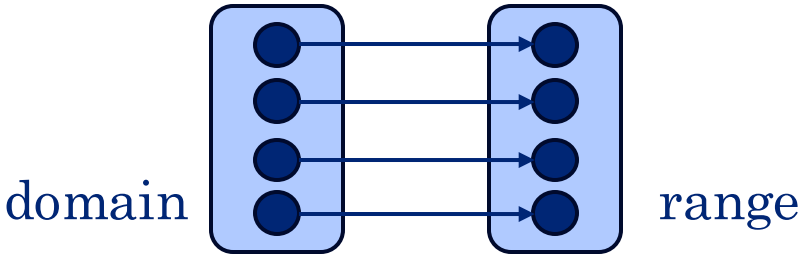
Every image has a label within our set of labels.

surjective – every output in our *range* maps to **at least one** input in *domain*



Every label is associated with at least one image.

bijective – there exists **exactly one** output in *range* for every input in *domain*



We have a perfect set of image-label pairs.

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Properties of Linear Functions

$$\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$$

vector of pixels

$$\text{dog} = f(\boldsymbol{x}) = \boldsymbol{A}\boldsymbol{x}$$

$$\boldsymbol{A} \in \mathbb{R}^{K \times D}$$

f is **injective** – if A is *full column rank* $\longrightarrow$ $\mathrm{R}(\boldsymbol{A}) = \mathbb{R}^D$ $\quad \mathrm{rank}(\boldsymbol{A}) = D$

f is **surjective** – if A is *full row rank* $\longrightarrow$ $\mathrm{R}(\boldsymbol{A}) = \mathbb{R}^K$ $\quad \mathrm{rank}(\boldsymbol{A}) = K$

f is **bijective** – if A is *full rank/invertible* $\longrightarrow$ $\mathrm{R}(\boldsymbol{A}) = \mathbb{R}^D$, $\mathrm{R}(\boldsymbol{A}) = \mathbb{R}^K$ $\quad \mathrm{rank}(\boldsymbol{A}) = D = K$

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Determining Matrix Rank

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 2 & 4 & 6 \end{bmatrix} \in \mathbb{R}^{3\times 3}$$

Option 1: Python

```python
import numpy as np

A = np.array([[1, 2, 3],
              [0, 1, 2],
              [2, 4, 6]])

print(np.linalg.matrix_rank(A))
```

✓ 0.0s

```
2
```

Option 2: Inspection

How many linearly independent rows/columns?

There are two; notice that:
row3 = 2 * row1
col3 = 2 * col2 – col1

Option 3: Singular Values

How many non-zero singular values?

Practice calculating singular values: Discussion 1 problem 2b

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Multivariate Calculus Review

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ML as Function Approximation

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vector of pixels

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$\boldsymbol{f}$?

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vector of words

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$\boldsymbol{g}$?

# Bed	# Bath	…	Location
4	3	…	Berkeley

$$\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$$

vector of features

$\$1{,}200{,}000 = h(\boldsymbol{x})$

$\boldsymbol{h}$?

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Sensitivity of Function

How sensitive is my output to my input variables?

How will my output change if I change my input by a small amount?

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix} \in \mathbb{R}^D$$

$$\mathrm{y} = f(\boldsymbol{x}) \in \mathbb{R}$$

If I change the first input variable (x_1) by a small amount (Δx_1) by how much will my output (y) change?

$$\Delta\, \mathrm{y} \approx \frac{\partial f}{\partial x_1}(\boldsymbol{x}) * \Delta x_1$$

partial derivative!

Practice calculating partial derivatives: Discussion 1 problem 1

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Learning New Features

$\xrightarrow{f_1\ ?}$ **Image Features** $\xrightarrow{f_2\ ?}$ **dog**

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$\xrightarrow{g_1\ ?}$ **Text Features** $\xrightarrow{g_2\ ?}$ **spam**

# Bed	# Bath	…	Location
4	3	…	Berkeley

$\xrightarrow{h_1\ ?}$ **Tabular Features** $\xrightarrow{h_2\ ?}$ **$1,200,000**

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Chain Rule

How sensitive is my output to my input variables?

How will my output change if I change my input by a small amount?

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix} \in \mathbb{R}^D$$

$$\mathrm{y} = f_2(f_1(\boldsymbol{x})) \in \mathbb{R}$$

If I change the first input variable (x_1) by a small amount (Δx_1) by how much will my output (y) change?

$$\Delta\, \mathrm{y} \approx \frac{\partial}{\partial x_1} f_2(f_1(\boldsymbol{x})) * \Delta x_1$$

$$\Delta\, \mathrm{y} \approx \frac{\partial f_2}{\partial x_1}(f_1(\boldsymbol{x})) * \frac{\partial f_1}{\partial x_1}(\boldsymbol{x}) * \Delta x_1$$

Practice with chain rule:
Discussion 1 problem 1a

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Probability Theory Review

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ML as Function Approximation

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vector of pixels

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vector of words

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vector of features

$\$1{,}200{,}000 = h(\boldsymbol{x})$

$\boldsymbol{h}$?

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Probabilistic Interpretation of ML

$$\boldsymbol{x} = \begin{bmatrix} 120 \\ 50 \\ 240 \\ \vdots \\ 15 \end{bmatrix} \in \mathbb{R}^D$$

vector of pixels

$P(\mathbf{Y} = \text{dog} \mid \boldsymbol{X} = \boldsymbol{x})$?

HI, It's your boss. Im stuck in Nigeria with none money. Please wire to TRWIGB2LXXX SOON.

$$\boldsymbol{x} = \begin{bmatrix} HI, \\ It's \\ \vdots \\ SOON. \end{bmatrix} \in \mathbb{R}^D$$

vector of words

$P(\mathbf{Y} = \text{spam} \mid \boldsymbol{X} = \boldsymbol{x})$?

# Bed	# Bath	…	Location
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$$\boldsymbol{x} = \begin{bmatrix} 4 \\ 3 \\ \vdots \\ Berkeley \end{bmatrix} \in \mathbb{R}^D$$

vector of features

$P(\mathbf{Y} = \$1.2 \text{ mil.} \mid \boldsymbol{X} = \boldsymbol{x})$?

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Bayes' Theorem

$$P(Y = k \mid X = x) = \frac{P(X=x \mid Y=k)P(Y=k)}{P(X=x)}$$

$$= \frac{P(X=x \mid Y=k)P(Y=k)}{\sum_y P(X=x|Y=y)P(Y=y)}$$

Law of total probability:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$= \sum_y P(X = x \mid Y = y)P(Y = y)$$

It might be hard to model $P(Y = k \mid X = x)$ directly. In some cases, it's easier to estimate $P(X = x|Y = y)$ and $P(Y = y)$, then use Bayes' theorem to estimate $P(Y = k \mid X = x)$.

Practice with Bayes' Theorem: Discussion 1 problem 3b

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Discussion Mini Lecture 1

Math for Machine Learning

Contributors: Sara Pohland

Additional Resources

1. ML as Function Approximation
   - Lecture 1 and 3
2. Linear Algebra
   - Deep Learning Foundations and Concepts – Appendix A
   - Mathematics for Machine Learning – Section 3
3. Multivariate Calculus
   - Calculus Early Transcendentals – Chapter 14
   - Mathematics for Machine Learning – Section 4
4. Probability Theory
   - Lecture 4
   - Deep Learning Foundations and Concepts – Chapter 2
   - Mathematics for Machine Learning – Section 5

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